Establish strong duality for distorted optimal transport beyond affine costs

`W4412630514_p0` (website page)

Website #23 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

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Problem

Let

$$P_{h,c}(\mu,\nu) := \inf\{\mathbb{E}_h[c(X,Y)] : X \sim \mu,\ Y \sim \nu\},$$

and

$$D_{h,c}(\mu,\nu) := \sup_{(\varphi,\psi)\in\mathcal{S}_c} \Big(\mathbb{E}_h[\varphi(X_\mu)] + \mathbb{E}_h[\psi(X_\nu)]\Big), \qquad \mathcal{S}_c := \{(\varphi,\psi) : \varphi(x)+\psi(y) \le c(x,y)\ \forall x,y\}.$$

The question asks for conditions on a distortion h and a measurable cost $c : \mathbb{R}^2 \to \mathbb{R}$ under which

$$P_{h,c}(\mu,\nu) = D_{h,c}(\mu,\nu) \qquad \text{for all probability measures } \mu,\nu \text{ on } \mathbb{R},$$

and in particular whether such strong duality can hold beyond the affine-cost case.

Alternative readings considered

There is a genuine boundedness ambiguity: the prompt defines $\mathcal{X}$ as bounded random variables, but then quantifies over all $\mu,\nu \in \mathcal{P}(\mathbb{R})$, for which $c(X,Y)$ need not be bounded. I therefore adopt the strongest coherent reading:

- positive results are stated for bounded Borel costs c, which makes both sides well-defined for all μ,ν;
- negative results are allowed to use unbounded costs only when the chosen marginals have bounded support, so all random variables appearing in the example are still bounded.

This is logically stronger than restricting the measures.

Related literature found

- **Haiyan Liu, Bin Wang, Ruodu Wang, Sheng Chao Zhuang, *Distorted optimal transport*** — **PARTIAL**. This is the exact paper behind the prompt. It introduces distorted OT, proves universal-optimality results for several classes of distortions/costs, and states a weak-duality theorem for convex distortions. However, its displayed affine-cost strong-duality statement cannot be correct as written for arbitrary h: its own Example 1 already points the other way, and below I give a short Bernoulli counterexample. Hence I used the paper for context, not as a proof tool. (sas.uwaterloo.ca)
- **Ruodu Wang, Yunran Wei, Gordon E. Willmot, *Characterization, Robustness, and Aggregation of Signed Choquet Integrals*** — **TOOL**. I use its quantile representation and its equivalence "concave distortion $\iff$ convex-order consistency $\iff$ subadditivity" for signed Choquet integrals. Via the dual distortion $\tilde{h}(t) = 1 - h(1-t)$, this yields the convex-order monotonicity and superadditivity needed for convex distortions. (pubsonline.informs.org)

- **R. Kaas, J. Dhaene, D. Vyncke, M. Goovaerts, M. Denuit,** ***A Simple Geometric Proof that Comonotonic Risks Have the Convex-Largest Sum*** — **TOOL**. I use that among couplings with fixed marginals, the comonotonic sum is maximal in convex order, and that quantiles add under comonotonicity. (cambridge.org)
- **Amir Dembo,** ***Probability Theory: STAT310/MATH230*** — **TOOL**. I use the existence of regular conditional probability distributions for real-valued random variables in order to lift couplings of pushforward marginals $f_{\#}\mu$ and $g_{\#}\nu$ back to couplings of μ, ν. (web.stanford.edu)
- **Mathias Beiglböck, Christian Léonard, Walter Schachermayer,** ***A general duality theorem for the Monge–Kantorovich transport problem*** — **TOOL**. I use the bounded-cost Kellerer corollary for the classical branch $h(t) = t$. (mat.univie.ac.at)

Proof sketch

There are three logically separate points.

First, for the classical linear distortion $h(t) = t$, one falls back to ordinary Kantorovich duality. That gives a large positive class of costs.

Second, for genuinely nonlinear distortions there is still a nontrivial positive class beyond affine costs: bounded *separable* costs

$$c(x, y) = f(x) + g(y).$$

The right way to look at this is to push μ, ν forward by f, g, reduce the primal to a distorted transport problem for the linear cost $u + v$, and then couple the transformed marginals comonotonically. For convex h, the dual distortion $\tilde{h}(t) = 1 - h(1 - t)$ is concave; using known properties of concave Choquet integrals and the identity

$$\mathbb{E}_h[Z] = -\mathbb{E}_{\tilde{h}}[-Z],$$

one gets that $\mathbb{E}_h$ is *superadditive* and *decreasing in convex order*. Since the comonotonic sum is convex-largest, it minimizes $\mathbb{E}_h[U + V]$. Quantile additivity for comonotonic sums then gives

$$\inf \mathbb{E}_h[U + V] = \mathbb{E}_h[U] + \mathbb{E}_h[V].$$

This matches the dual value with the obvious dual pair (f, g).

Third, this is close to the limit of what one can hope for in general. I prove two obstructions.

- Even for the affine cost $x + y$, if one asks for strong duality for *all* marginals, then necessarily

  $$h(p) + h(1 - p) \leq 1 \quad \forall p \in [0, 1].$$

  So affine-cost duality certainly does *not* hold for arbitrary distortions.

- For strictly convex h, a very simple bounded non-separable cost already exhibits a real duality gap:

  $$c_\lambda(x, y) = \mathbf{1}_{(0,\infty)}(x) + \mathbf{1}_{(0,\infty)}(y) + \lambda\, \mathbf{1}_{(0,\infty)}(x)\mathbf{1}_{(0,\infty)}(y), \qquad \lambda > 0.$$

  With Bernoulli marginals, the dual side is exactly $2h(1/2)$, while the primal side is strictly larger.

So the picture that emerges is: *yes*, strong duality survives beyond affine costs, but a robust class is "convex distortion + separable cost", not "arbitrary distortion + affine cost".

Main result

Theorem 1. *Let h be a distortion function.*

(i) If $h(t) = t$ and $c : \mathbb{R}^2 \to \mathbb{R}$ is bounded Borel measurable, then

$$P_{h,c}(\mu, \nu) = D_{h,c}(\mu, \nu) \qquad \text{for all } \mu, \nu \in \mathcal{P}(\mathbb{R}).$$

(ii) If h is convex and c is bounded and separable,

$$c(x, y) = f(x) + g(y) \qquad (f, g : \mathbb{R} \to \mathbb{R} \text{ bounded Borel}),$$

then for all $\mu, \nu \in \mathcal{P}(\mathbb{R})$,

$$P_{h,c}(\mu, \nu) = D_{h,c}(\mu, \nu) = \mathbb{E}_h[f(X_\mu)] + \mathbb{E}_h[g(X_\nu)].$$

(iii) A necessary condition for strong duality even for the single affine cost $c(x, y) = x + y$ to hold for all marginals is

$$h(p) + h(1 - p) \le 1 \qquad \forall p \in [0, 1].$$

Consequently, if $h(p) + h(1 - p) > 1$ for some p, then strong duality fails for the affine cost $x + y$. In particular, every non-linear concave distortion fails this test.

(iv) If h is strictly convex, then for every $\lambda > 0$ the bounded Borel cost

$$c_\lambda(x, y) := \mathbf{1}_{(0,\infty)}(x) + \mathbf{1}_{(0,\infty)}(y) + \lambda\, \mathbf{1}_{(0,\infty)}(x)\mathbf{1}_{(0,\infty)}(y)$$

does not *satisfy strong duality for all marginals. More precisely, for*

$$\mu = \nu = \tfrac{1}{2}\delta_0 + \tfrac{1}{2}\delta_1,$$

one has

$$D_{h,c_\lambda}(\mu, \nu) = 2h(1/2) \quad \text{but} \quad P_{h,c_\lambda}(\mu, \nu) > 2h(1/2).$$

Proof

For convenience, define the dual distortion

$$\tilde{h}(t) := 1 - h(1 - t), \qquad t \in [0, 1].$$

Lemma 1. *For every bounded random variable Z,*

$$\mathbb{E}_h[Z] = -\mathbb{E}_{\tilde{h}}[-Z].$$

Proof. Write

$$\mathbb{E}_{\tilde{h}}[-Z] = \int_0^\infty \tilde{h}(\mathbb{P}(-Z > x))\, dx + \int_{-\infty}^0 \big(\tilde{h}(\mathbb{P}(-Z > x)) - 1\big)\, dx.$$

Using $\mathbb{P}(-Z > x) = \mathbb{P}(Z < -x)$ and the changes of variables $y = -x$ in both integrals,

$$-\mathbb{E}_{\tilde{h}}[-Z] = \int_{-\infty}^0 \big(1 - \tilde{h}(\mathbb{P}(Z < y))\big)\, dy + \int_0^\infty \big(1 - \tilde{h}(\mathbb{P}(Z < y))\big)\, dy - \int_{-\infty}^0 dy.$$

Since $1 - \tilde{h}(t) = h(1-t)$, this becomes

$$-\mathbb{E}_{\tilde{h}}[-Z] = \int_0^\infty h\big(1 - \mathbb{P}(Z < y)\big)\, dy + \int_{-\infty}^0 \Big(h\big(1 - \mathbb{P}(Z < y)\big) - 1\Big)\, dy.$$

Now $1 - \mathbb{P}(Z < y) = \mathbb{P}(Z \geq y)$. For a real-valued random variable, the set of atoms

$$A := \{y \in \mathbb{R} : \mathbb{P}(Z = y) > 0\}$$

is countable; hence $\mathbb{P}(Z \geq y) = \mathbb{P}(Z > y)$ for Lebesgue-a.e. y. Replacing $\geq$ by $>$ in the last display therefore does not change either integral. The resulting expression is exactly $\mathbb{E}_h[Z]$. □

Lemma 2. *Suppose $f, g : \mathbb{R} \to \mathbb{R}$ are Borel and bounded. Let $\bar{\mu} = f_{\#}\mu$ and $\bar{\nu} = g_{\#}\nu$. Then every coupling $\pi \in \Pi(\bar{\mu}, \bar{\nu})$ can be lifted to some $\gamma \in \Pi(\mu, \nu)$ such that*

$$(f, g)_{\#}\gamma = \pi.$$

Consequently,

$$P_{h,f+g}(\mu, \nu) = \inf\{\mathbb{E}_h[U + V] : U \sim \bar{\mu},\ V \sim \bar{\nu}\}.$$

Proof. Take $X \sim \mu$ and put $U = f(X)$. By existence of regular conditional probability distributions for real-valued random variables, there exists a kernel $K(u, \cdot)$ giving a regular conditional law of X given $\sigma(U)$. Likewise, if $Y \sim \nu$ and $V = g(Y)$, there exists a kernel $L(v, \cdot)$ giving a regular conditional law of Y given $\sigma(V)$. Hypothesis audit: Dembo's Proposition 4.4.3 assumes only that the conditioned random variable is real-valued and the conditioning object is a σ-algebra; here X, Y, U, V are real-valued and $\sigma(U), \sigma(V)$ are σ-algebras, so the proposition applies. (web.stanford.edu)

Define γ on $\mathbb{R}^2$ by

$$\gamma(A \times B) := \int_{\mathbb{R}^2} K(u, A)\, L(v, B)\, \pi(du, dv).$$

Its first marginal is μ, because

$$\gamma(A \times \mathbb{R}) = \int K(u, A)\, \bar{\mu}(du) = \mu(A),$$

the last equality being the defining disintegration identity of the regular conditional law of X given U. Similarly the second marginal is ν.

It remains to check $(f, g)_{\#}\gamma = \pi$. For Borel $C, D \subseteq \mathbb{R}$,

$$(f, g)_{\#}\gamma(C \times D) = \int K(u, f^{-1}(C))\, L(v, g^{-1}(D))\, \pi(du, dv).$$

Since

$$\mathbf{1}_{\{U \in C\}} = \mathbf{1}_{\{X \in f^{-1}(C)\}} \quad \text{a.s.},$$

taking conditional expectation given $\sigma(U)$ yields

$$K(U, f^{-1}(C)) = \mathbf{1}_{\{U \in C\}} \quad \text{a.s.},$$

hence $K(u, f^{-1}(C)) = \mathbf{1}_C(u)$ for $\bar{\mu}$-a.e. u. Likewise $L(v, g^{-1}(D)) = \mathbf{1}_D(v)$ for $\bar{\nu}$-a.e. v. Because π has marginals $\bar{\mu}, \bar{\nu}$, the integral reduces to

$$\int \mathbf{1}_C(u) \mathbf{1}_D(v)\, \pi(du, dv) = \pi(C \times D).$$

So $(f, g)_{\#}\gamma = \pi$.

The final identity follows immediately: every coupling (X, Y) of (μ, ν) induces a coupling $(f(X), g(Y))$ of $(\bar{\mu}, \bar{\nu})$, and every coupling of $(\bar{\mu}, \bar{\nu})$ lifts back. □

Lemma 3. *If h is convex, then:*

(a) $\mathbb{E}_h$ *is superadditive:*

$$\mathbb{E}_h[U+V] \geq \mathbb{E}_h[U] + \mathbb{E}_h[V] \qquad (U, V \in \mathcal{X}).$$

(b) $\mathbb{E}_h$ *is decreasing in convex order:*

$$U \leq_{cx} V \implies \mathbb{E}_h[U] \geq \mathbb{E}_h[V].$$

Proof. If h is convex, then $\tilde{h}$ is concave. For concave distortions, signed/ordinary Choquet integrals are subadditive and increasing in convex order by Wang–Wei–Willmot, Theorem 3. Hypothesis audit: their theorem is stated for signed Choquet integrals I_g on L^∞ with $g \in H$; our $\tilde{h}$ is a bounded-variation function on $[0,1]$ and our variables are bounded, so the theorem applies. Their I_g uses $\mathbb{P}(X \geq x)$ rather than $\mathbb{P}(X > x)$, but for bounded real-valued X the two versions coincide because atoms occur at only countably many levels, hence the Lebesgue integrals agree. (math.uwaterloo.ca)

For (a),

$$\mathbb{E}_h[U+V] = -\mathbb{E}_{\tilde{h}}[-U-V] \geq -\mathbb{E}_{\tilde{h}}[-U] - \mathbb{E}_{\tilde{h}}[-V] = \mathbb{E}_h[U] + \mathbb{E}_h[V],$$

using the previous lemma and subadditivity of $\mathbb{E}_{\tilde{h}}$.

For (b), if $U \leq_{cx} V$, then also $-U \leq_{cx} -V$ because composition with $x \mapsto -x$ preserves convexity of test functions. Since $\tilde{h}$ is concave, $\mathbb{E}_{\tilde{h}}$ is increasing in convex order, so

$$\mathbb{E}_{\tilde{h}}[-U] \leq \mathbb{E}_{\tilde{h}}[-V].$$

Multiplying by -1 and using the sign-change lemma gives

$$\mathbb{E}_h[U] \geq \mathbb{E}_h[V]. \qquad \square$$

Proof of the theorem. **Part (i).** If $h(t) = t$, then $\mathbb{E}_h$ is the ordinary expectation. For bounded Borel c, let $m := \inf c$ and $\bar{c} := c - m \geq 0$. By Corollary 2.4 of Beiglböck–Léonard–Schachermayer, the classical Monge–Kantorovich duality has no gap for bounded nonnegative Borel costs on Polish spaces. Hypothesis audit: $X = Y = \mathbb{R}$ are Polish, μ, ν are Borel probabilities, and $\bar{c}$ is bounded Borel and nonnegative. So

$$\inf_{\pi \in \Pi(\mu,\nu)} \int \bar{c}\, d\pi = \sup_{\varphi + \psi \leq \bar{c}} \left(\int \varphi\, d\mu + \int \psi\, d\nu \right).$$

Adding back m yields the same equality for c. (mat.univie.ac.at)

Part (ii). Fix bounded Borel f, g and write $c = f + g$. Set $\bar{\mu} = f_{\#}\mu$, $\bar{\nu} = g_{\#}\nu$. By the lifting lemma,

$$P_{h,c}(\mu,\nu) = \inf\{\mathbb{E}_h[U+V] : U \sim \bar{\mu},\ V \sim \bar{\nu}\}.$$

Let $W \sim U[0,1]$ on an atomless space and define the comonotonic coupling

$$U^c := Q_{\bar{\mu}}(W), \qquad V^c := Q_{\bar{\nu}}(W).$$

By Kaas–Dhaene–Vyncke–Goovaerts–Denuit, Theorem 6, for every other coupling $U \sim \bar{\mu}$, $V \sim \bar{\nu}$,

$$U + V \leq_{cx} U^c + V^c.$$

Hypothesis audit: their theorem applies to random vectors with prescribed marginals; we use it for $n = 2$, with (U^c, V^c) comonotonic and sharing the marginals of (U, V). Therefore Lemma 3(b) gives

$$\mathbb{E}_h[U^c + V^c] \leq \mathbb{E}_h[U + V]$$

for every coupling, so

$$P_{h,c}(\mu, \nu) = \mathbb{E}_h[U^c + V^c].$$

Next, convexity of h implies continuity on $[0, 1]$, hence left-continuity. By Wang–Wei–Willmot, Lemma 3(ii),

$$\mathbb{E}_h[Z] = \int_0^1 Q_{1-p}(Z)\, dh(p)$$

for bounded Z. By Kaas et al., Theorem 7, quantiles add under comonotonicity:

$$Q_q(U^c + V^c) = Q_q(U^c) + Q_q(V^c) \qquad (q \in (0, 1)).$$

Hypothesis audit: Theorem 7 assumes only comonotonicity of the vector, which holds by construction. Therefore

$$\mathbb{E}_h[U^c + V^c] = \int_0^1 \big(Q_{1-p}(U^c) + Q_{1-p}(V^c)\big)\, dh(p) = \mathbb{E}_h[U^c] + \mathbb{E}_h[V^c].$$

Since $\mathbb{E}_h$ is law invariant and $U^c \sim \bar{\mu} = f(X_\mu)$, $V^c \sim \bar{\nu} = g(X_\nu)$,

$$P_{h,c}(\mu, \nu) = \mathbb{E}_h[f(X_\mu)] + \mathbb{E}_h[g(X_\nu)].$$

The cited quantile and convex-order facts are exactly the tools used in this paragraph. (cambridge.org)

It remains to match the dual value. Let $(\varphi, \psi) \in \mathcal{S}_c$. For any coupling $X \sim \mu$, $Y \sim \nu$,

$$c(X, Y) \geq \varphi(X) + \psi(Y) \quad \text{a.s.}$$

Hence by monotonicity of $\mathbb{E}_h$ and Lemma 3(a),

$$\mathbb{E}_h[c(X, Y)] \geq \mathbb{E}_h[\varphi(X) + \psi(Y)] \geq \mathbb{E}_h[\varphi(X)] + \mathbb{E}_h[\psi(Y)].$$

Taking the infimum over X, Y and then the supremum over $(\varphi, \psi) \in \mathcal{S}_c$,

$$P_{h,c}(\mu, \nu) \geq D_{h,c}(\mu, \nu).$$

But $(f, g) \in \mathcal{S}_c$ because $f(x) + g(y) = c(x, y)$. Therefore

$$D_{h,c}(\mu, \nu) \geq \mathbb{E}_h[f(X_\mu)] + \mathbb{E}_h[g(X_\nu)] = P_{h,c}(\mu, \nu).$$

So equality holds throughout:

$$P_{h,c}(\mu, \nu) = D_{h,c}(\mu, \nu) = \mathbb{E}_h[f(X_\mu)] + \mathbb{E}_h[g(X_\nu)].$$

Part (iii). Fix $p \in [0, 1]$. Let

$$\mu = (1 - p)\delta_0 + p\delta_1, \qquad \nu = p\delta_0 + (1 - p)\delta_1.$$

Take $X \sim \mu$ and define $Y := 1 - X$. Then $Y \sim \nu$ and

$$X + Y \equiv 1.$$

Hence

$$P_{h,x+y}(\mu, \nu) \leq \mathbb{E}_h[1] = 1.$$

Now choose the feasible dual pair $\varphi(x) = x$, $\psi(y) = y$, for which $\varphi + \psi = x + y = c$. Since X_μ is Bernoulli with success probability p,

$$\mathbb{E}_h[\varphi(X_\mu)] = \mathbb{E}_h[X_\mu] = h(p).$$

Likewise,

$$\mathbb{E}_h[\psi(X_\nu)] = h(1 - p).$$

Therefore

$$D_{h,x+y}(\mu, \nu) \geq h(p) + h(1 - p).$$

If strong duality held for this cost and these marginals, we would have

$$h(p) + h(1 - p) \leq D_{h,x+y}(\mu, \nu) = P_{h,x+y}(\mu, \nu) \leq 1.$$

Thus $h(p) + h(1 - p) \leq 1$ is necessary for every p.

If h is concave and non-linear, then $h(t) \geq t$ for all $t \in [0, 1]$, with strict inequality for some $t \in (0, 1)$; hence $h(p)+h(1-p) > 1$ for some p. So strong duality fails already for the affine cost $x+y$. This directly contradicts the "for all h" affine statement in Liu et al. as written. (sas.uwaterloo.ca)

Part (iv). Fix $\lambda > 0$ and set

$$c_\lambda(x, y) = \mathbf{1}_{(0,\infty)}(x) + \mathbf{1}_{(0,\infty)}(y) + \lambda\, \mathbf{1}_{(0,\infty)}(x)\mathbf{1}_{(0,\infty)}(y).$$

Let $\mu = \nu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$, and abbreviate

$$k := h(1/2).$$

Strict convexity of h and the endpoint conditions $h(0) = 0$, $h(1) = 1$ imply

$$k < 1/2.$$

Primal lower bound. Any coupling of μ, ν is determined by

$$t := \mathbb{P}(X = 1, Y = 1) \in [0, 1/2].$$

Then automatically

$$\mathbb{P}(X = 0, Y = 0) = t, \qquad \mathbb{P}(X \neq Y) = 1 - 2t.$$

So $Z := c_\lambda(X, Y)$ satisfies

$$\mathbb{P}(Z = 0) = t, \qquad \mathbb{P}(Z = 1) = 1 - 2t, \qquad \mathbb{P}(Z = 2 + \lambda) = t.$$

Therefore

$$\mathbb{E}_h[Z] = h(1 - t) + (1 + \lambda)h(t) =: F_\lambda(t).$$

If $t = 0$, then $F_\lambda(0) = 1 > 2k$. If $0 < t < 1/2$, strict convexity gives

$$h(t) + h(1 - t) > 2h(1/2) = 2k,$$

hence

$$F_\lambda(t) = h(1-t) + h(t) + \lambda h(t) > 2k.$$

If $t = 1/2$, then

$$F_\lambda(1/2) = 2k + \lambda k > 2k.$$

So

$$P_{h,c_\lambda}(\mu, \nu) = \inf_{t \in [0,1/2]} F_\lambda(t) > 2k.$$

Dual exact value. Restrict a feasible pair $(\varphi, \psi) \in \mathcal{S}_{c_\lambda}$ to $\{0, 1\}$. Write

$$a_i := \varphi(i), \qquad b_j := \psi(j) \qquad (i, j \in \{0, 1\}).$$

Because replacing (φ, ψ) by $(\varphi - \alpha, \psi + \alpha)$ leaves both feasibility and the dual objective unchanged, we may assume $a_0 = 0$. Put

$$u := a_1, \qquad v := b_0, \qquad w := b_1.$$

Feasibility gives

$$v \leq 0, \qquad w \leq 1, \qquad u + v \leq 1, \qquad u + w \leq 2 + \lambda.$$

For a Bernoulli-1/2 random variable taking values r, s, one checks directly that

$$\mathbb{E}_h[Z] = \begin{cases} r + k(s-r), & s \geq r, \\ r + (1-k)(s-r), & s < r. \end{cases}$$

Define

$$R(z) := \begin{cases} kz, & z \geq 0, \\ (1-k)z, & z < 0. \end{cases}$$

Then

$$\mathbb{E}_h[\varphi(X_\mu)] = R(u), \qquad \mathbb{E}_h[\psi(X_\nu)] = v + R(w - v).$$

Now $u \leq 1 - v$. Hence

$$R(u) \leq k(1-v).$$

Indeed, if $u \geq 0$ this is immediate from $R(u) = ku$; if $u < 0$, then $R(u) < 0 \leq k(1-v)$. Likewise $w - v \leq 1 - v$ because $w \leq 1$, and therefore

$$R(w - v) \leq k(1 - v)$$

by the same case split. Consequently

$$\mathbb{E}_h[\varphi(X_\mu)] + \mathbb{E}_h[\psi(X_\nu)] \leq 2k + (1-2k)v \leq 2k$$

since $v \leq 0$ and $k < 1/2$. So

$$D_{h,c_\lambda}(\mu, \nu) \leq 2k.$$

On the other hand, the pair

$$\varphi(x) = \mathbf{1}_{(0,\infty)}(x), \qquad \psi(y) = \mathbf{1}_{(0,\infty)}(y)$$

is feasible because $\varphi + \psi \leq c_\lambda$, and it gives

$$\mathbb{E}_h[\varphi(X_\mu)] + \mathbb{E}_h[\psi(X_\nu)] = k + k = 2k.$$

Hence

$$D_{h,c_\lambda}(\mu, \nu) = 2k.$$

Combining the primal and dual estimates,

$$D_{h,c_\lambda}(\mu, \nu) = 2h(1/2) < P_{h,c_\lambda}(\mu, \nu).$$

So strong duality fails. $\square$

Gap to the full statement

What remains unproved is a full characterization.

More precisely, I have shown:

- a broad positive class: $h = \mathrm{id}$ with bounded Borel c, and convex h with bounded separable $c = f + g$;
- a necessary condition on h already for the affine cost $x + y$: $h(p) + h(1 - p) \leq 1$ for all p;
- a genuine non-separable obstruction for strictly convex h.

I have *not* characterized all distortions h for which strong duality holds for the affine cost $x+y$, nor all bounded non-separable costs c for convex but not strictly convex h, nor the correct unbounded integrability regime.

Self red-team

- **Counterexample attempts against part (ii).** I tried to break the separable-cost theorem with non-monotone f, g. The initial monotone-coupling proof was too narrow, but the lifting lemma reduces the problem to couplings of $f(X)$ and $g(Y)$, so monotonicity of f, g is actually unnecessary.
- **Boundary checks.**
  - If $h(t) = t$, part (iv) must collapse; it does: then $k = 1/2$, the dual value is 1, and the primal infimum is also 1 (attained at $t = 0$).
  - If $\lambda = 0$ in part (iv), the cost becomes separable, so the gap should disappear; it does.
  - If $p = 0$ or 1 in part (iii), the necessary condition gives equality $1 \leq 1$, as it should.
- **Three likely referee objections.**
  1. *"You used results for Choquet integrals defined with $\mathbb{P}(X \geq x)$, but the problem uses $\mathbb{P}(X > x)$."* I addressed this explicitly: for bounded real random variables the exceptional levels are atoms of the distribution, hence countable, so the Lebesgue integrals coincide.
  2. *"The lifting lemma may fail because the prescribed coupling of $f(X)$ and $g(Y)$ need not come from a coupling of X and Y."* This is exactly why the proof uses regular conditional distributions of X given $f(X)$ and of Y given $g(Y)$; the verification that $(f, g)_{\#}\gamma = \pi$ is written out in full.
  3. *"Maybe Liu et al.'s affine theorem had an implicit extra assumption."* The displayed theorem in the paper does not state one, and the Bernoulli example in part (iii) lies entirely in the bounded setting of the paper. So as written, the "all h" affine claim cannot stand.
- **Degenerate cases checked.** Dirac marginals, constant costs, and one-variable costs $c(x, y) = f(x)$ all reduce correctly.

Ideas tried

1. **Direct verification of the paper's affine theorem.** I first tried to reproduce the claimed affine-cost duality for arbitrary h. The obstruction appeared immediately: for $c(x, y) = x + y$ and a constant-sum coupling $Y = 1 - X$, the primal is at most 1, while the dual pair $(\varphi, \psi) = (x, y)$ gives $h(p) + h(1 - p)$. This produced the necessary condition in part (iii) and exposed the affine-all-h claim as untenable.

2. **Attempted full 2×2 characterization of all non-separable costs.** I reduced general counterexamples to binary marginals and wrote the primal as a one-parameter family depending on the 2×2 coupling parameter t. This did yield the explicit non-separable gap in part (iv), but I could not force a gap for *every* nonlinear convex h: piecewise-linear convex distortions such as $h(t) = (2t - 1)_+$ make the most natural interaction example collapse back to equality.

3. **Monotone-separable proof via quantile couplings.** A first positive proof worked for monotone f, g by choosing comonotonic or countermonotonic couplings of X, Y. The obstruction was that it did not cover general measurable f, g. The natural next step was to lift couplings of the pushforward marginals $f_{\#}\mu$ and $g_{\#}\nu$; regular conditional probabilities remove that obstruction and yield the stronger bounded-separable theorem proved above.

4. **Using universal-optimal copulas for inverse-S / S-shaped distortions.** I explored whether the primal optimal-coupling results in Liu et al. could be turned into strong duality. The obstruction is conceptual: knowing the primal optimizer is not enough. One still needs a separable minorant $\varphi + \psi$ saturated on the optimizer *and* an additivity mechanism on the dual side. Outside the separable/comonotonic setting, I could not make those two requirements meet.

Invoked results

1. **Bounded-cost classical Kantorovich duality** (**TOOL**). *Statement.* Let X, Y be Polish spaces with Borel probability measures μ, ν, and let $c : X \times Y \to [0, \infty]$ be Borel measurable and uniformly bounded. Then there is no duality gap: the primal optimal transport value equals the supremum over measurable potentials φ, ψ with $\varphi + \psi \le c$. This is Corollary 2.4 in Beiglböck–Léonard–Schachermayer, citing Kellerer. *Citation.* Beiglböck, Léonard, Schachermayer (2012), *Studia Math.* 209, 151–167, DOI 10.4064/sm209-2-4. (mat.univie.ac.at) *Hypothesis audit.* In part (i), $X = Y = \mathbb{R}$ are Polish, μ, ν are Borel probabilities, and after shifting by $m = \inf c$, the cost $c - m$ is bounded, Borel, and nonnegative.

2. **Existence of regular conditional probability distributions for real-valued random variables** (**TOOL**). *Statement.* For any real-valued random variable X and any sub-σ-algebra $\mathcal{G}$, there exists a regular conditional probability distribution of X given $\mathcal{G}$. *Citation.* Dembo (2019), *Probability Theory: STAT310/MATH230*, Proposition 4.4.3. (web.stanford.edu) *Hypothesis audit.* In the lifting lemma, X, Y, $f(X)$, and $g(Y)$ are all real-valued; the conditioning σ-algebras are $\sigma(f(X))$ and $\sigma(g(Y))$.

3. **Comonotonic sums are convex-largest** (**TOOL**). *Statement.* If $(Y_1, \dots, Y_n)$ is comonotonic and has the same marginals as $(X_1, \dots, X_n)$, then

$$X_1 + \cdots + X_n \le_{cx} Y_1 + \cdots + Y_n.$$

Citation. Kaas, Dhaene, Vyncke, Goovaerts, Denuit (2002), Theorem 6, *ASTIN Bulletin* 32(1), 71–80, DOI 10.2143/AST.32.1.1015. (cambridge.org) *Hypothesis audit.* In part (ii) we apply it with $n = 2$, $Y_1 = U^c$, $Y_2 = V^c$ comonotonic, and $X_1 = U$, $X_2 = V$ arbitrary with the same marginals.

4. **Quantiles add for comonotonic sums (TOOL).** *Statement.* If $(Y_1, \ldots, Y_n)$ is comonotonic, then each u-quantile of $Y_1 + \cdots + Y_n$ equals the sum of the u-quantiles of the Y_i. *Citation.* Kaas, Dhaene, Vyncke, Goovaerts, Denuit (2002), Theorem 7. (cambridge.org) *Hypothesis audit.* In part (ii), U^c and V^c are comonotonic by construction, so their quantiles add.

5. **Concave distortions are exactly the convex-order consistent and subadditive Choquet integrals (TOOL).** *Statement.* For $h \in H$, the following are equivalent: (i) h is concave; (ii) I_h is convex-order consistent; (iii) I_h is subadditive; (and several further equivalent properties). *Citation.* Wang, Wei, Willmot (2020), Theorem 3, *Mathematics of Operations Research* 45(3), 993–1015, DOI 10.1287/moor.2019.1020. (math.uwaterloo.ca) *Hypothesis audit.* We apply this to $\tilde{h}$. Since h is a distortion, $\tilde{h}(t) = 1 - h(1-t)$ is of bounded variation and belongs to H. When h is convex, $\tilde{h}$ is concave. Our variables are bounded, so $I_{\tilde{h}}$ is well-defined. As noted in the proof, $I_{\tilde{h}}$ coincides with $\mathbb{E}_{\tilde{h}}$ for bounded real variables.

6. **Quantile representation for left-continuous h (TOOL).** *Statement.* If $h \in H$ is left-continuous, then
$$I_h(X) = \int_0^1 F_X^{-1}(1-p)\, dh(p)$$
for bounded X. *Citation.* Wang, Wei, Willmot (2020), Lemma 3(ii). (math.uwaterloo.ca) *Hypothesis audit.* In part (ii), h is convex, hence continuous on $[0, 1]$, so in particular left-continuous; U^c, V^c are bounded because f, g are bounded.

Self-confidence

3

I checked every step of the stated partial results carefully, including the Bernoulli counterexamples and the separable-cost proof, but I did not obtain a full characterization of all admissible (h, c).

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